TEXORA · LECTURE SHEET

Printing Machinery

Printing is localised dyeing done by machinery: each printing machine is a way of placing a measured film of paste at a measured position, thousands of times a minute. Roller, flat screen, rotary screen and digital machines divide the industry's work between them.

WP-3207 · ADVANCED MACHINERIES AND RELEVANT CALCULATIONS · TOPIC 8 · SPECIALIZATION — WET PROCESSING

1. What every printing machine must do

A design is separated into one **colourway element per colour**; each colour needs its own printing unit (roller, screen or print head set), and the units must print in **register** — exact positional agreement — on a fabric moving at production speed. After printing, the fabric is dried, the colour is fixed (steaming or curing), and unfixed colour and thickener are washed off. Machines differ in how the paste is transferred and in the length of design repeat they allow.

2. Roller printing — the engraved cylinder

Each colour is an engraved copper roller; paste is fed to its surface, a doctor blade wipes the land clean, and the engraved cells deliver paste to the fabric pressed against a central pressure bowl. Strengths: very fine detail, long runs at high speed, excellent repeatability. Weaknesses: engraving is expensive and slow, so short runs and frequent design changes are uneconomic; the repeat is limited by roller circumference. Roller printing survives where long runs of classic designs justify the engraving cost.

3. Flat-bed screen printing — the versatile batch-continuous hybrid

A flat screen (nylon mesh on a frame) is laid on the fabric; a squeegee drives paste through the open mesh; the screen lifts, the fabric indexes forward by one repeat, and the cycle repeats along a long table or an automatic carriage machine. Strengths: large repeats possible, heavy paste deposits (good for pigments and special effects), modest screen cost. Weaknesses: slow compared with rotary; joins at each index can show on critical designs. Flat-bed remains the choice for large-repeat furnishing designs, panels (placement prints) and sampling.

4. Rotary screen printing — the industry standard

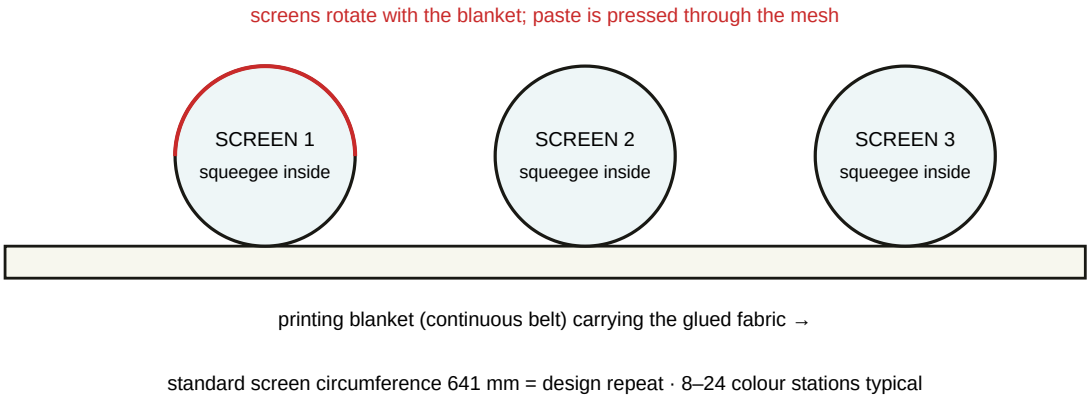


Fig. 1 — Rotary screen printing (schematic): seamless nickel screens rotate over a moving blanket; each station prints one colour of the design in continuous register.

The screen is a seamless perforated nickel cylinder. Paste is pumped inside; a stationary squeegee presses it through the mesh as the screen rotates in contact with the fabric on a continuous blanket. Because printing is continuous, speeds of 40–100 m/min are routine, with the repeat fixed by the screen circumference — the standard 641 mm (also 820, 914, 1018 mm). Register is held by individual drives per station. Rotary screen dominates world print production because it balances speed, repeat economy and design versatility.

WORKED EXAMPLE 1 — ROTARY OUTPUT AND REPEAT FITTING

Machine speed 60 m/min; fabric 1.6 m wide, 200 g/m² (0.20 kg/m²).

Throughput = 60 × 60 × 1.6 × 0.20 = **1,152 kg/h** (5,760 m/h).

Repeat fitting: the design repeat must divide the screen circumference exactly. A 320.5 mm design repeat fits the 641 mm screen exactly twice per revolution (641/320.5 = 2.00) ✓; a 300 mm repeat does not (641/300 = 2.14) and must be re-scaled to 320.5 mm or moved to a different screen circumference. Repeat arithmetic is checked at design stage, not at the machine.

5. Digital (inkjet) printing — no screens at all

Piezoelectric print heads jet low-viscosity inks (reactive, acid, disperse, pigment, sublimation) directly onto pre-treated fabric, drop on demand, from a digital file. Strengths: no screens or engraving, photographic detail, unlimited colours per design, economical from the first metre — sampling and short runs transformed, and mass customisation possible. Weaknesses: lower maximum speed than rotary, ink and pre-treatment costs per metre higher, penetration and fastness depend on the full pre-treat → print → steam → wash chain being correct. Digital is growing fastest precisely where rotary is weakest (Topic 11).

6. Auxiliary machinery of the print works

- **Screen making:** coating, exposure and washing units for flat screens; laser engraving for rotary nickel screens.
- **Colour kitchen:** automatic dispensing and mixing of pastes by weight to the recipe (WP-4101 Topic 4), with viscosity control — paste rheology is a printing variable.
- **Steamer:** loop or star steamer giving saturated-steam dwell for fixation after printing (reactive prints: about 102 °C saturated steam).
- **Washing-off range:** open-width soaping and washing to remove thickener and unfixed dye without staining the white ground.
- **Sample and strike-off machines:** short tables or mini-rotary units proving the design before production.

7. Machine selection summary

MACHINE	REPEAT LIMIT	SPEED CHARACTER	BEST WORK
Roller	Roller circumference	Very fast, long runs	Fine classic designs in volume
Flat-bed screen	Screen length (large)	Slow, indexed	Large repeats, panels, heavy deposits
Rotary screen	641 mm standard	Fast, continuous	General apparel and furnishing production
Digital inkjet	None (file-based)	Slowest per metre, zero set-up	Samples, short runs, photographic designs

8. Glossary — key terms

- **Register:** exact positional agreement of successive colours in a design.
- **Repeat:** the length of fabric after which the design pattern repeats.
- **Squeegee:** the blade that presses paste through a screen mesh.
- **Blanket:** the endless belt carrying fabric through a rotary printer.
- **Strike-off:** a short proof print of a new design for approval.

9. Key points

- The four machine families divide work by repeat, run length and deposit: roller for volume classics, flat-bed for size, rotary for general production, digital for freedom from screens.
- Rotary repeat must divide the screen circumference exactly — checked in the design office (641 = 2 × 320.5).
- Printing machinery includes its chain: screen making, colour kitchen, steamer and washing-off decide quality as much as the printer.

10. Summary

A print works is a system for placing paste in register at speed: 1,152 kg/h through the rotary of Fig. 1, each of its screens a colour of the design, each metre afterwards steamed and washed. Machine choice follows the design's repeat, the run's length and the deposit the effect demands — the same engineering matching of machine to goods as in dyeing.

11. Practice

1. A rotary printer runs 45 m/min on 1.5 m wide fabric of 180 g/m². Calculate output in m/h and kg/h.
2. A design has a 273 mm repeat. Does it fit a 641 mm screen? What circumferences from 820, 914 and 1018 mm fit it as an integer number of repeats within 1%? State the best choice.
3. Give two reasons digital printing wins sampling work even though its running cost per metre is higher than rotary.

Answer key

1. 45 × 60 = 2,700 m/h. Mass = 2,700 × 1.5 × 0.180 = 729 kg/h.
2. 641/273 = 2.35 — no. 820/273 = 3.004 ≈ 3 repeats (error 0.1%) ✓ best choice; 914/273 = 3.35, 1018/273 = 3.73 — no.
3. No screens to make (cost and days saved — a design change is a new file, not new hardware), and the first metre costs the same as any other, so a one-metre proof carries no set-up penalty; photographic colour range is also unlimited by station count.